Bayesian Marketing Mix Prototype

Goal: learn how marketing affects synthetic sales and brand, then forecast and plan budgets.

Built and tested: 156 weeks x 4 products x 3 price tiers (1,872 observations); 130 training weeks and 26 held-out weeks. Baseline, stockout-only, funnel-only, and combined variants passed predefined gates; 19 deterministic regression tests passed.

Methods implemented: Bayesian partial pooling (shrinkage), ad carryover, search funnel, Kalman-filtered brand scores, and profit/CVaR budget optimization.

MODEL: WHAT IT REPRESENTS

Demand and hierarchy. Product, tier, and product x tier baseline effects partially pool. Controls: price, macro, base-price growth, promotion, post-promotion dip, PrimeDays, and annual seasonality. No random-walk baseline.

Media shrinkage. Positive product responses partially pool toward learned channel centers. Display, Video, Social, and CTV have estimated adstock and saturation; paid clicks have no adstock. Social is S-shaped. Tier slopes and transforms remain shared; no-pooling superiority is untested.

Funnel. CTV, organic demand, and controls drive branded search. Search has log-Normal measurement noise; clicks are Binomial with spend-dependent capture. Sales use latent mean clicks, not observed holdout outcomes; CTV also has a direct effect.

Kalman brand state. A jointly estimated 0-100 score follows a linear-Gaussian logit state process with shocks and noisy score measurements. A Kalman filter updates states from training scores; nonlinear sales are not Kalman observations.

New business inputs. Sponsorship has impressions and separate spend at a synthetic $18 CPM. Weekly product distribution is the volume-weighted % of available stores/market carrying the product, not store count. It is fixed in planning; no separate SKU-breadth variable.

Sales and decisions. Censored log-Normal sales = min(demand, inventory). Six spend lines include sponsorship. Compare units, expected contribution-profit, and 50/50 expected/5%-CVaR objectives over 13 weeks. Margin rates are illustrative, not observed costs; budget and product proportions are fixed. CTV attribution avoids double counting.

COMBINED RUN: EVIDENCE

Sampling reliability
R-hat 1.0055; 0 divergences

Holdout log-sales error
87.3% lower RMSE*

90% predictive coverage
Demand 90.1%; sales 92.6%

Funnel coverage / CTV
Search 91.3%; clicks 94.2%; CTV lift error 2.8%

Known media-effect recovery
20/20 product intervals; max error 4.39%

Brand score holdout
RMSE 1.26 points; 87.5% coverage

Persistence / baseline demand
0.904 weekly; +8.5% per +10 prior score pts

5% profit CVaR
$944.3m baseline; $952.8m risk allocation

13-week decisions
$7.48m; units +0.54%; profit +$5.1m

*Versus a product-tier mean log-sales benchmark. All gates passed. Budget conserved; unit-objective shortfall is below 0.0001% versus the known-truth best found under the same constraints. Six local starts are not a global-optimum proof.

NEXT STEPS: MMM BEST PRACTICE AND QUANT RESEARCH

1. Identification. Use geo/lift evidence; address spend endogeneity, mediation confounding, brand construct validity, and physical distribution effects.

2. Robustness. Repeat seeds, untouched/rolling holdouts, simulation-based calibration, prior sensitivity, and misspecification stress tests.

3. Real-data pilots. Audit score surveys, exposures, cost, sales, and weighted distribution; specify plausible future demand, supply, and coverage scenarios.

4. Quant decisions. Add profit and posterior tail-loss (CVaR) constraints; research longer-horizon brand value, pacing, uncertain costs, and replenishment.

5. Dynamic models. Test richer brand shocks, nonlinear state filters, time-varying response and lag forms, and joint brand/supply dynamics; require decision value.

Boundary: one deliberately informative matched synthetic seed; not independent real-data evidence. Future inputs/coverage/inventory and margins are supplied assumptions. Kalman filtering uses scores, not nonlinear sales observations. No direct distribution constraint or separate assortment breadth. Attribution is a convention; 13 weeks omit later brand value. **Not real-world causal validation or deployment advice.**

Alternative Methods: Compare and Contrast

A frequentist Ridge model is a useful benchmark, but it is not a drop-in equivalent to this Bayesian hierarchical MMM.

BAYESIAN HIERARCHICAL MMM

What it means. Parameters are represented by probability distributions updated with observed data and prior assumptions. Posterior draws describe uncertainty about effects and predictions, conditional on the model and data.

What this prototype adds. Product effects partially pool toward shared channel centers. The model also represents nonlinear media response, a search/click funnel, censored sales, persistent brand state, and uncertainty for downstream decisions.

Costs and caveats. MCMC takes longer and requires convergence diagnostics. Results depend on model structure and priors; posterior uncertainty does not by itself establish causal effects or protect against misspecification.

FREQUENTIST RIDGE REGRESSION

What it means. Coefficients are treated as fixed but unknown. Inference asks how estimates or procedures would vary across repeated samples. Ridge fits coefficients with an L2 penalty, usually tuned by time-aware validation.

What it buys. Fast fitting, stable estimates under correlated predictors, and a simple predictive benchmark. It can show whether the more elaborate model materially improves held-out forecasts.

What it gives up. Ordinary Ridge shrinks toward zero, not toward a learned population effect; hierarchy needs extra structure. It gives point estimates, so uncertainty needs methods such as bootstrap. Standard Ridge also allows negative coefficients unless constrained.

AT A GLANCE

	Bayesian model	Ridge
Pooling	Bayesian model: explicit partial pooling across products.	Ridge: global shrinkage; group-specific pooling must be designed.
Uncertainty	Bayesian model: posterior and predictive scenarios feed profit/CVaR.	Ridge: bootstrap intervals and financial scenarios need additional work.
Model structure	Bayesian model: supports the prototype's funnel, censoring, and brand state.	Ridge: nonlinear features can be engineered, but these mechanisms need separate treatment.
Best fit	Bayesian model: richer inference and uncertainty-aware decisions.	Ridge: rapid baseline, prediction benchmark, and simpler repeated refits.

A FAIR TEST FOR THIS MMM

Fit Ridge to the same training window and inputs, using time-based cross-validation to choose its penalty. Keep all feature construction and scaling training-only; compare both models on the same untouched holdout for forecast error and calibration. Then compare attribution and optimized budgets: similar RMSE alone does not mean the decisions agree.

Practical recommendation: use the Bayesian model to compare unit, expected-profit, and CVaR allocations; replace illustrative margins with real costs before business use, and add Ridge as a fast predictive benchmark. In a linear Gaussian model with fixed Gaussian priors, Ridge corresponds to a Bayesian MAP estimate, but not the full posterior uncertainty or this prototype's hierarchical structure.